

L12a: How Do Micro Decisions Lead to Emergent Macro Trends?

CHEME 5660 Cornell University

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Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

Today, we write down simple rules for individual traders, simulate many of them together, and measure what the market as a whole does.

Today's concepts

- Agent-based models: the five parts every model shares, in the minority game and the Farmer–Joshi market.
- The Boltzmann choice rule: derived from maximum entropy, with a temperature that sets how strongly agents favor high scores.
- Micro to macro: volatility in the minority game; fat tails and volatility clustering in the Farmer–Joshi market.

What You Need to Know

Core concepts

- Five parts: agents, information, a decision rule, an aggregate, and feedback from the aggregate to the next decisions.
- Maximum entropy for a fixed expected score gives $\pi_s = \exp(U_s/T)/\mathcal{Z}$: greedy as $T \to 0$, uniform as $T \to \infty$.
- The minority game's volatility per agent depends on α and T; Farmer–Joshi value investors and trend followers together produce fat tails and volatility clustering.

For understanding (you won't reproduce these)

- The critical value $\alpha_c \approx 0.3374$ (cited, not derived) and $-d\mathcal{H}/d\bar{U} = 1/T$.
- The Farmer–Joshi threshold rules and our parameter values.

Lectures and Examples

Lecture:

CHEME-5660-L12a-Lecture-AgentBasedMarkets-Fall-2026.ipynb

Examples:

CHEME-5660-L12a-Example-ThermalMinorityGame-Fall-2026.ipynb

CHEME-5660-L12a-Example-FarmerJoshi-Fall-2026.ipynb

The Structure of an Agent-Based Model

An agent-based model describes individual decision makers and their rules, simulates them together, and measures the market-wide result:

Part	Minority game	Farmer–Joshi market
Agents	N agents, S fixed strategies each	Value investors, trend followers
Information	History $\mu(t)$ of the last m outcomes	Mispricing or price change over a lag
Decision rule	Choose a strategy by its score	Enter or exit at a threshold
Aggregate	Attendance $A(t)$	Net order, then the price
Feedback	$A(t)$ updates scores and history	New price changes every signal

The Minority Game

Each round, N (odd) agents buy ($+1$) or sell (-1); the minority side, $-\text{sign}\, A(t)$, wins. Agent i plays the action $a_i(t)$ of one of its S strategies, tables a_{is}^{μ} over the $P = 2^m$ histories μ:

$$A(t) = \sum_{i=1}^{N} a_i(t) \qquad\qquad U_{is}(t+1) = U_{is}(t) - \frac{a_{is}^{\mu(t)} A(t)}{P}$$

where $A(t)$ is the attendance and $U_{is}(t)$ the score of strategy s of agent i. A strategy that would have put its agent in the minority gains score.

Source: Challet and Zhang (1997); scores divided by P as in Challet, Marsili, and Zecchina (2000)

Volatility per Agent

We measure how far the attendance moves from an even split, compared with N agents who flip fair coins (variance N):

$$\frac{\sigma^2}{N} = \frac{\langle A^2 \rangle}{N} \qquad \alpha = \frac{P}{N}$$

where $\langle \cdot \rangle$ averages over rounds and α is the number of histories per agent. $\sigma^2/N > 1$: more agents on the same side than with coin flips; $\sigma^2/N < 1$: a more even split. When α is small, many agents switch to the same action at the same time.

Choosing a Strategy: Maximum Entropy

An agent chooses strategy s with probability π_s. The entropy $\mathcal{H}(\pi)$ measures how spread out the choice is: zero for a certain choice, $\ln S$ for a uniform one. For a fixed expected score $\bar{U}$, we choose the π_s that commit to the least:

$$\max_{\pi} \; \mathcal{H}(\pi) = -\sum_{s=1}^{S} \pi_s \ln \pi_s \quad \text{subject to} \quad \sum_{s=1}^{S} \pi_s = 1, \qquad \sum_{s=1}^{S} \pi_s U_s = \bar{U}$$

with $\bar{U}$ strictly between the average and the largest score.

Source: Cavagna, Garrahan, Giardina, and Sherrington (1999)

The Boltzmann Rule

With Lagrange multipliers ϕ_0 (normalization) and ϕ_1 (expected score), setting $\partial \mathcal{L} / \partial \pi_s = 0$ gives:

$$-\ln \pi_s - 1 - \phi_0 + \phi_1 U_s = 0 \quad \Longrightarrow \quad \boxed{\pi_s = \frac{1}{\mathcal{Z}} \exp\left(\frac{U_s}{T}\right)}, \qquad \mathcal{Z} = \sum_{k=1}^{S} \exp\left(\frac{U_k}{T}\right)$$

where $T = 1/\phi_1$ is the temperature, in the units of the scores, and the normalizing sum $\mathcal{Z}$ is the partition function. The second derivative, $-1/\pi_s$, is negative, so this is a maximum.

What the Temperature Does

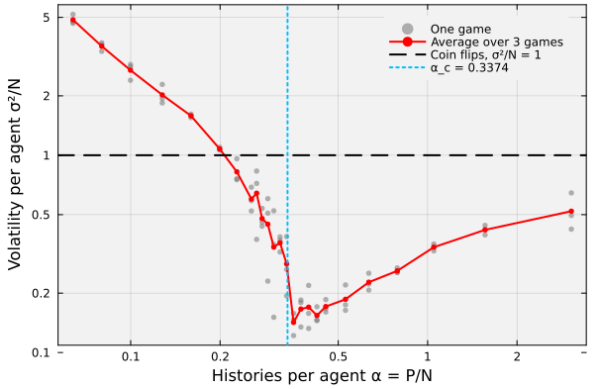
The temperature sets how much the choice depends on the scores:

- $T \to 0$: the best strategy has probability 1, the greedy minority game; with one best strategy, the entropy goes to zero.
- $T \to \infty$: every strategy has probability $1/S$; the entropy is $\ln S$.
- In between: with two scores Δ apart, the better strategy is played with probability $e^{\Delta/T}/(e^{\Delta/T}+1)$, about 0.73 at $T = \Delta$.

Physics: $-d\mathcal{H}/d\bar{U} = \phi_1 = 1/T$, the way thermodynamics defines temperature through entropy and energy.

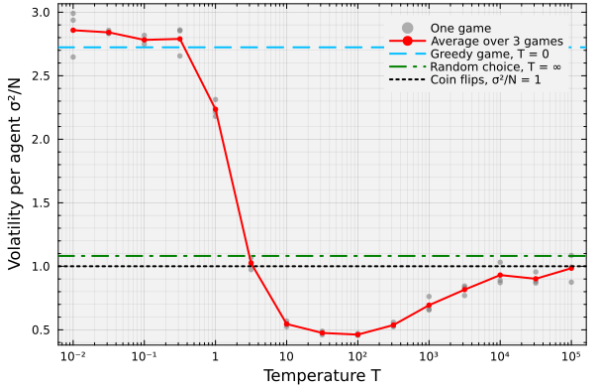
The Phase Diagram

Volatility per agent against α at $T = 0$ ($m = 6$, $S = 2$, three games per N). The minimum, 0.14 at $\alpha \approx 0.35$, is close to $\alpha_c \approx 0.3374$:



The Effect of Temperature

At $\alpha \approx 0.1$: greedy 2.72, random choice 1.08, and 0.46 at $T = 100$, below both limits. Our scores are divided by P, so our T is $1/\alpha$ times that of Cavagna and coauthors:



The Farmer–Joshi Market

Trader i holds $x_t^{(i)} \in \{-c_i, 0, +c_i\}$ and orders $\omega_{t+1}^{(i)} = x_{t+1}^{(i)} - x_t^{(i)}$. The market maker moves the log price in proportion to the net order:

$$p_{t+1} = p_t + \frac{1}{\lambda} \sum_i \omega_{t+1}^{(i)} + \xi_{t+1}$$

where λ is the liquidity and ξ_{t+1} is Normal noise with standard deviation σ_ξ. The log return is $r_{t+1} = p_{t+1} - p_t$.

Source: Farmer and Joshi (2002)

Value Investors and Trend Followers

Each trader enters at a signal beyond $\pm\chi_i^{\text{in}}$ and exits back across $\pm\chi_i^{\text{out}}$, with position size $c_i = a(\chi_i^{\text{in}} - \chi_i^{\text{out}})$:

- Value investors: signal $p_t - (v_t + v^{(i)})$; sell when overpriced, buy when underpriced.
- Trend followers: signal $p_t - p_{t-\theta_i}$; buy after a rise, sell after a fall.

Our variant: Table 1 traders, 1,200 of each, with $\sigma_\xi = 0.05$ (Table 1 lists 0.35, which lets the price drift far from the value) and a random-walk log value v_t.

Stylized Facts: Simulated and Real

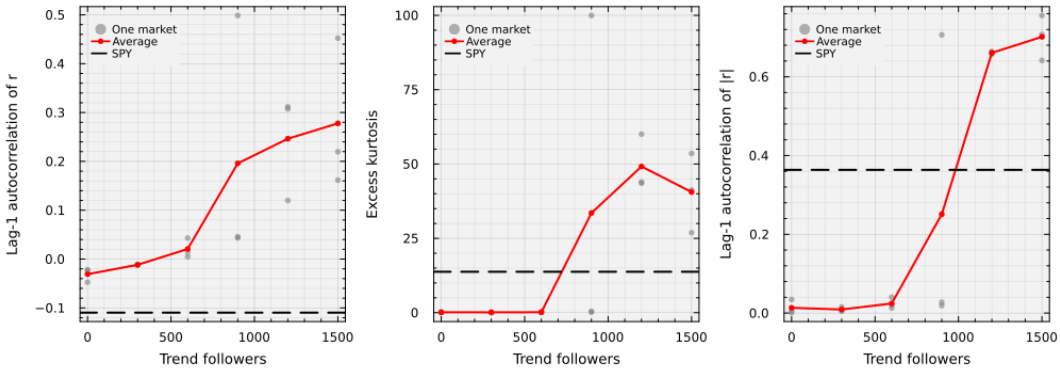
Both groups together, against `SPY` daily growth rates, 2014–2024 (kurtosis and autocorrelations do not depend on units):

Statistic	Simulated	`SPY`
Excess kurtosis	45.3	13.8
Autocorrelation of r, lag 1	0.27	-0.11
Autocorrelation of $\lvert r\rvert$, lag 1	0.67	0.36
Autocorrelation of $\lvert r\rvert$, lag 5	0.03	0.33
Autocorrelation of volume, lag 1	0.56	0.75

Fat tails and clustering come out of the rules, but the clustering fades within a few steps.

The Mix of Traders

With 1,200 value investors: kurtosis and clustering near zero with 600 or fewer trend followers; both large with 1,200 or more. Every mix misses `SPY` by a wide margin on at least one statistic:



Summary

We introduced agent-based models, derived the Boltzmann choice rule from maximum entropy, and connected trading rules to market-wide behavior.

Key takeaways

- Agent-based models: agents, information, a decision rule, an aggregate, and feedback.
- Maximum entropy: gives $\pi_s \propto \exp(U_s/T)$; the multiplier on the expected score is $1/T$.
- Micro to macro: the minority game's volatility depends on α and T; Farmer–Joshi traders produce fat tails and clustering, but not every stylized fact at once.